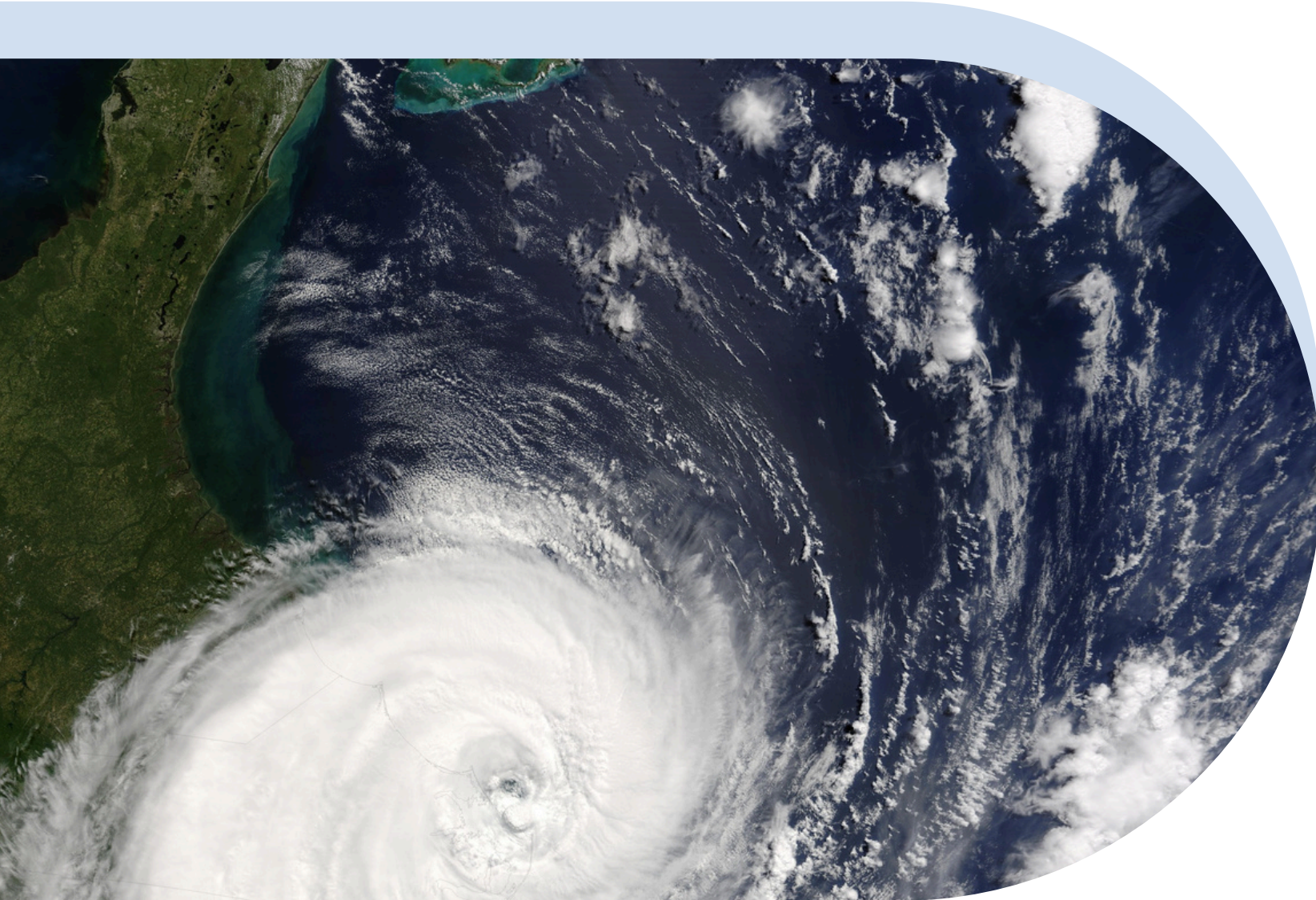




PRECISION RELIEF

**AI-Driven Command Intelligence
for Optimised Post-Flood
Recovery**

BE THIRD YEAR- COE/COPC

CPG Number: 174

Under the mentorship of: Mandeep Kaur

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(102317165) KANISHKA RANI

(102303126) VIVEK DUTT SHARMA

(102353004) PIYUSH GARG

PROJECT DEFINITION

- The project is an AI-based decision-support system that uses satellite and geospatial data to identify flooded areas and assess their impact on affected communities.
- It converts this information into prioritized and safer relief plans by identifying vulnerable areas, avoiding flooded roads, and optimizing routes for emergency resource delivery.

Project Scope

- The project covers end-to-end flood intelligence, from satellite-based flood detection and impact assessment to identifying vulnerable and isolated areas.
- It focuses on intelligent relief planning by detecting unsafe roads, prioritizing affected locations, estimating relief requirements, and generating optimized routes for emergency response.

PROJECT OBJECTIVES

• To develop and train the UNet and IBM-NASA Prithvi models to accurately detect flooded areas from satellite images and generate flood masks

1

• To estimate the number of people affected by the flood by combining the generated flood masks with population and district boundary data.

2

• To calculate flood severity and assign priority scores to districts so that the most affected areas can be identified for rescue and relief operations.

3

• To develop an interactive dashboard that displays flood maps, affected population, priority districts, and optimized rescue routes to support disaster management and decision-making.

4

Project Analysis

01

The analysis focuses on the limitations of traditional flood-response methods, such as delayed ground surveys, cloud-dependent imagery, static road maps, and manual relief coordination.

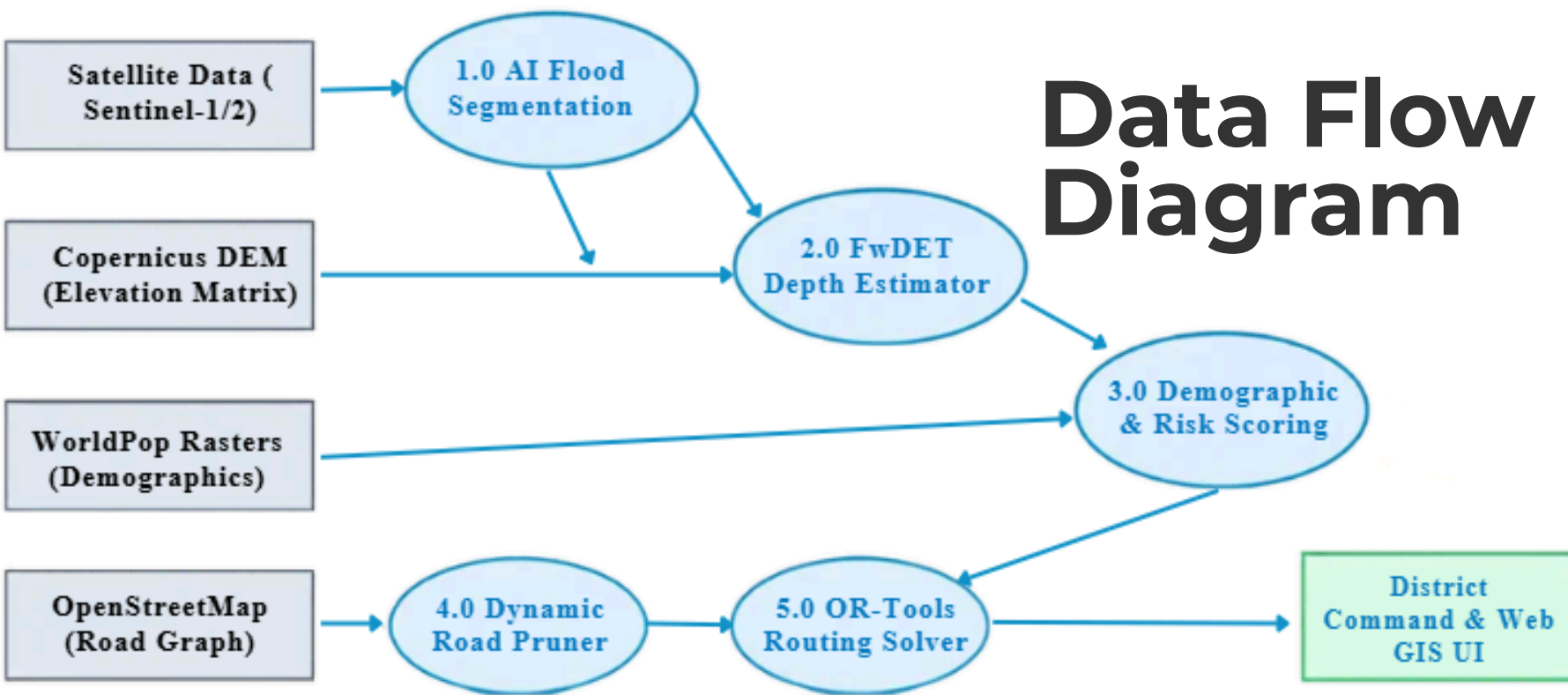
02

The proposed system analyzes flood extent, population impact, road accessibility, and vulnerability together, enabling the identification of high-priority areas and supporting data-driven relief decisions.

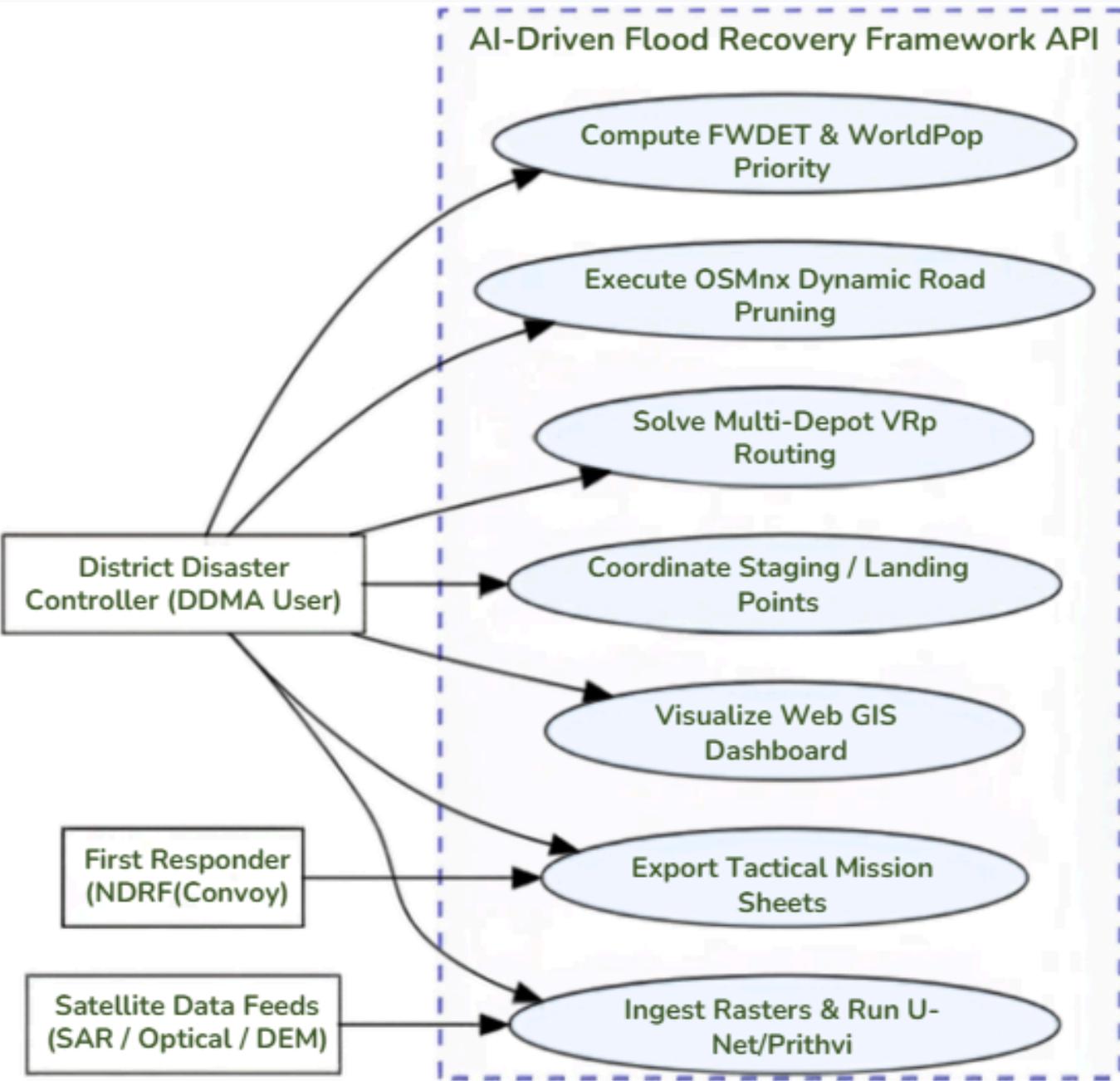
Literature Survey

TITLE	EXISTING METHODOLOGY	RESEARCH GAP
Two-stage distributionally robust optimization for disaster relief logistics under option contract and demand ambiguity	Robust optimization (DRO) to handle demand uncertainty and supply contracts.	Focused on pre-disaster inventory rather than tactical field navigation.
Sen1Floods11: a georeferenced dataset to train and test deep learning flood algorithms for SAR	Used FCNNs to segment surface water using SAR radar data.	Primarily focused on pixel-level classification (Flood vs No-Flood).
SAR-based flood mapping: a fully automated processing chain	Automated processing chain for NRT (Near-Real-Time) flood detection.	Lacked integration with urban street networks for rescue planning.
OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks	Tool for constructing and analyzing complex street networks from OSM.	Focus was on urban design, not on real-time disaster "Pruning."

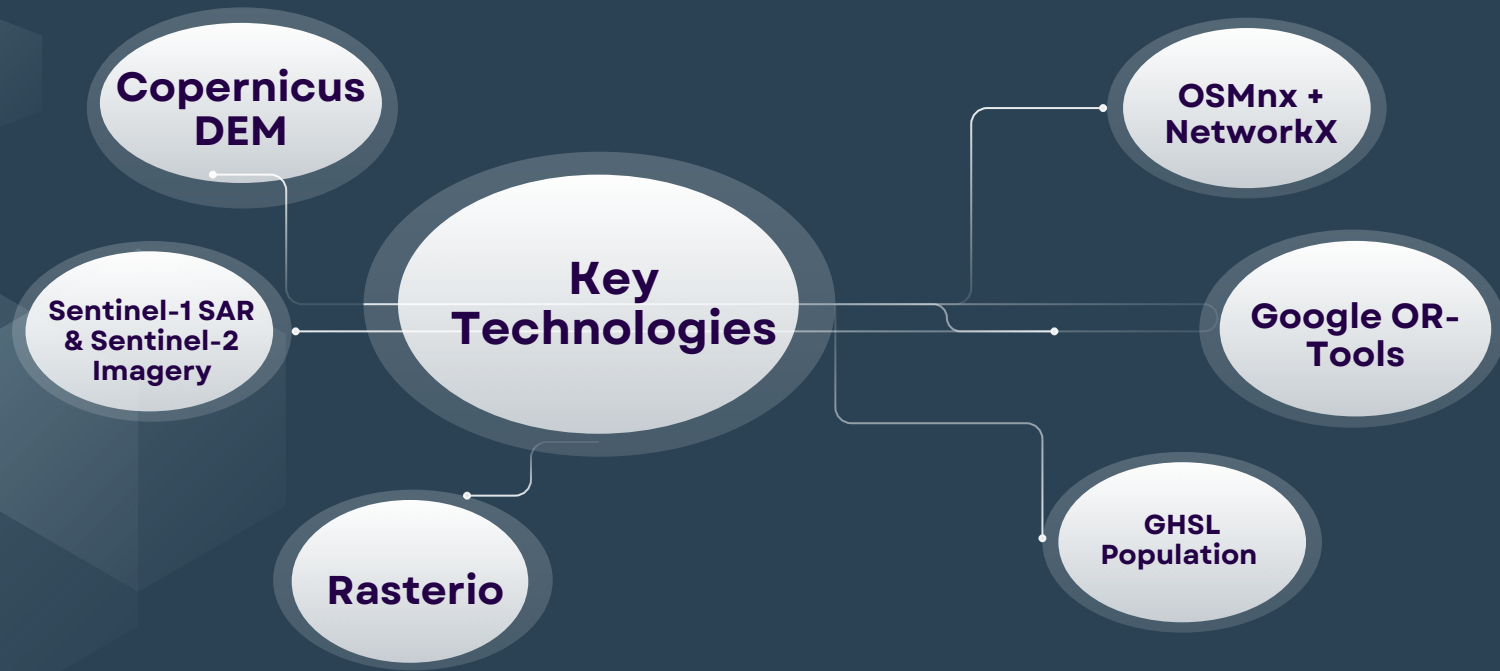
Data Flow Diagram



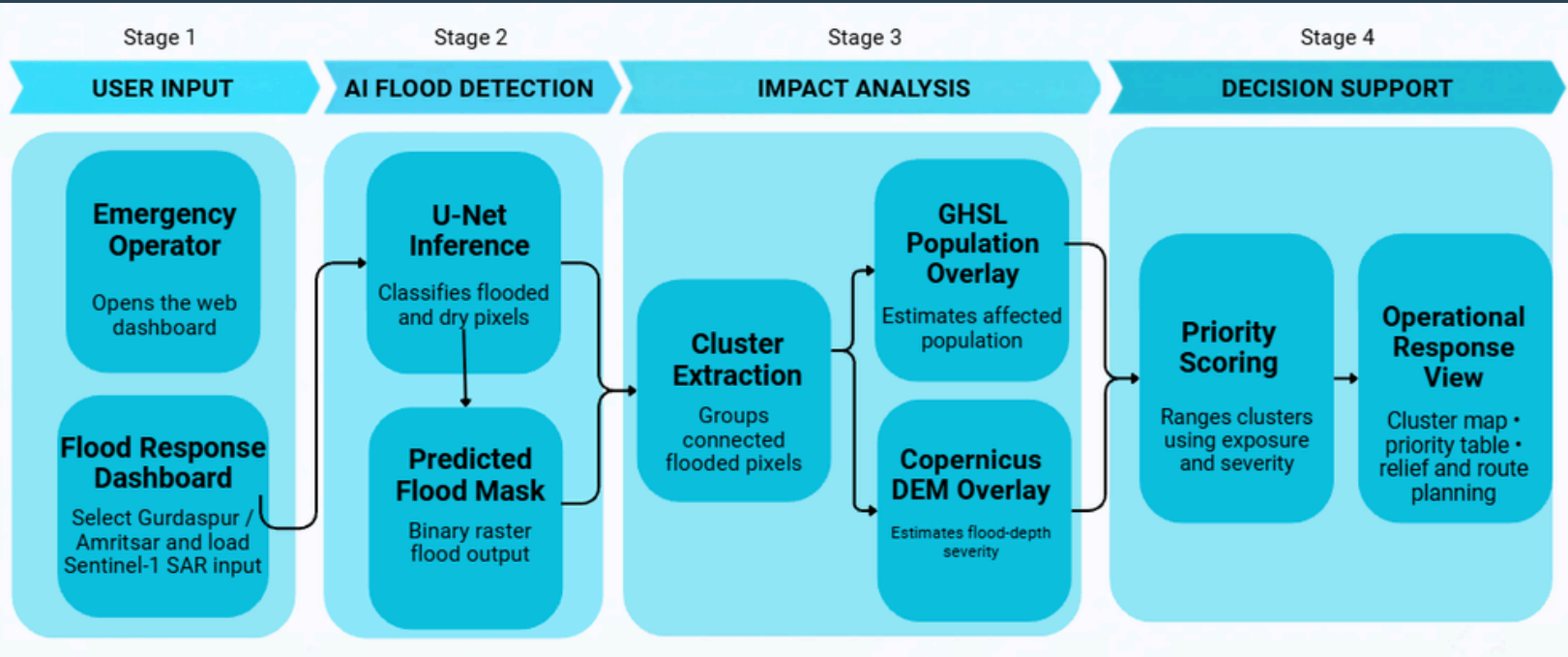
UseCase Diagram



Data Design



User Interface



1. SENSING & DATA INGESTION TIER

Sentinel-1 SAR Detects floods through clouds.	Sentinel-2 Optical Provides clear land images.	Copernicus DEM Measures land elevation and depth.	OpenStreetMap Provides road-network data	GHSL Population Shows population distribution.
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2. CORE AI & GEOSPATIAL INTELLIGENCE TIER

Deep Learning Segmentation • Custom U-Net (SAR Flood) • IBM-NASA Prithvi GeoFM • Adaptive Raster Fusion • Polygon Mask Vectorization	Priority-Scoring • FwDet Terrain Depth • Vector Cluster Exposure • GHSL Punjab Population • Multi-Factor Priority Scoring	Dynamic Routing Engine • OSMnx Road Graph Builder • Flooded Edge Pruning • Google OR:Tools • Senior authority board Discovery
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3. PRESENTATION & OPERATIONAL

Interactive Web GIS Dashboard Displays maps and key results.	Relief Mission Sheets Provides rescue-route instructions.
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ARCHITECTURAL DIAGRAM

Objective 1 Outcome

The Prithvi model uses Sentinel-2 optical imagery to identify flooded areas from surface features. It achieved 93.07% Recall and 89.29% Precision when compared with the reference flood mask.

Validation chip: India_900498 | Flood IoU: 67.8% | Flood F1: 80.8%

Sentinel-2 optical input (RGB)



Reference flood mask

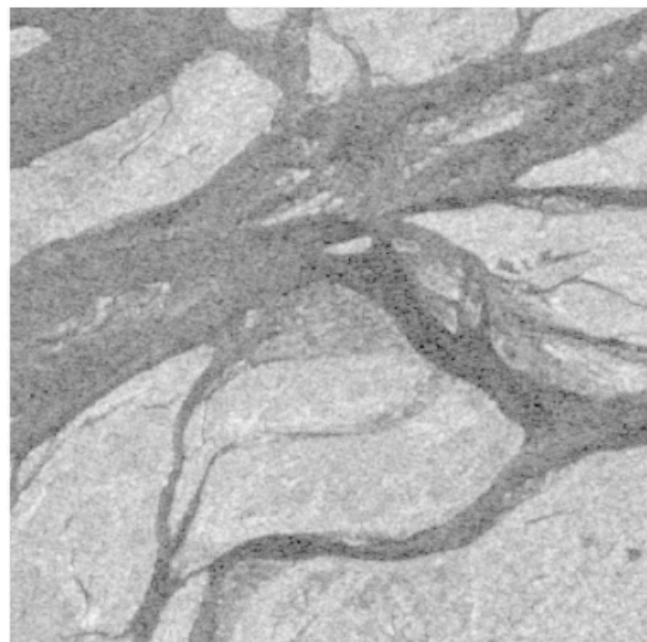


Prithvi flood prediction



Validation chip: India_900498 | Flood IoU: 76.8% | Flood F1: 86.9%

Sentinel-1 SAR input (VV)



Reference flood mask



U-Net flood prediction

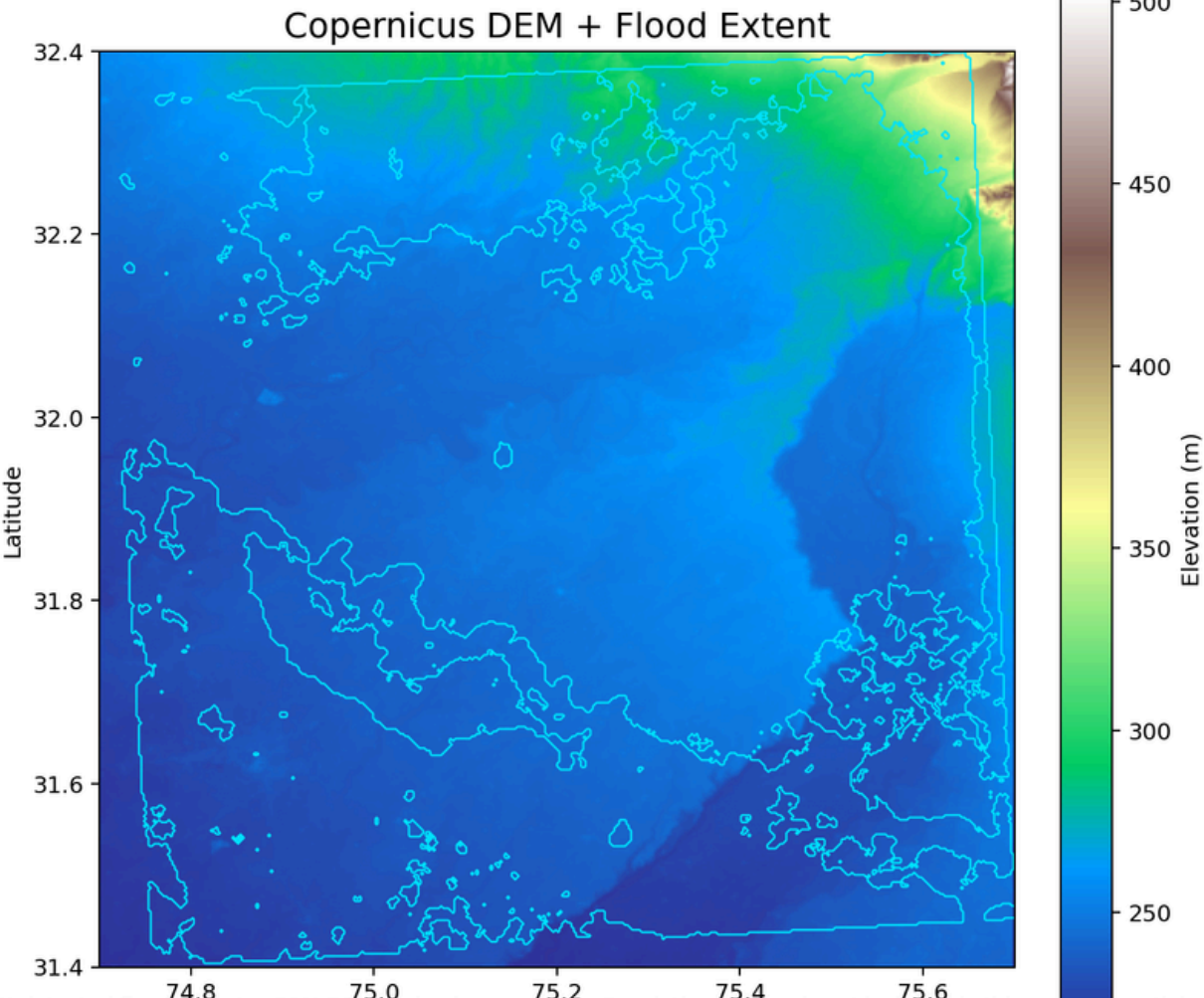
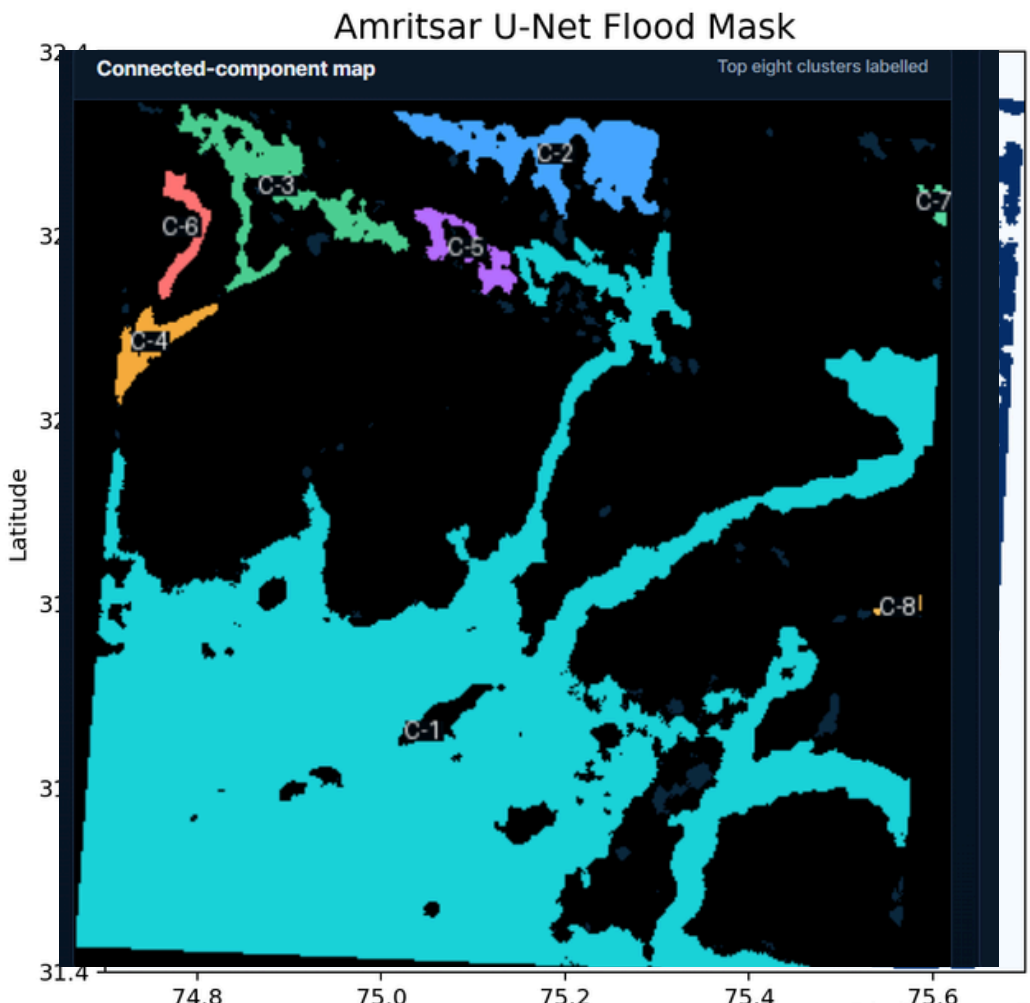
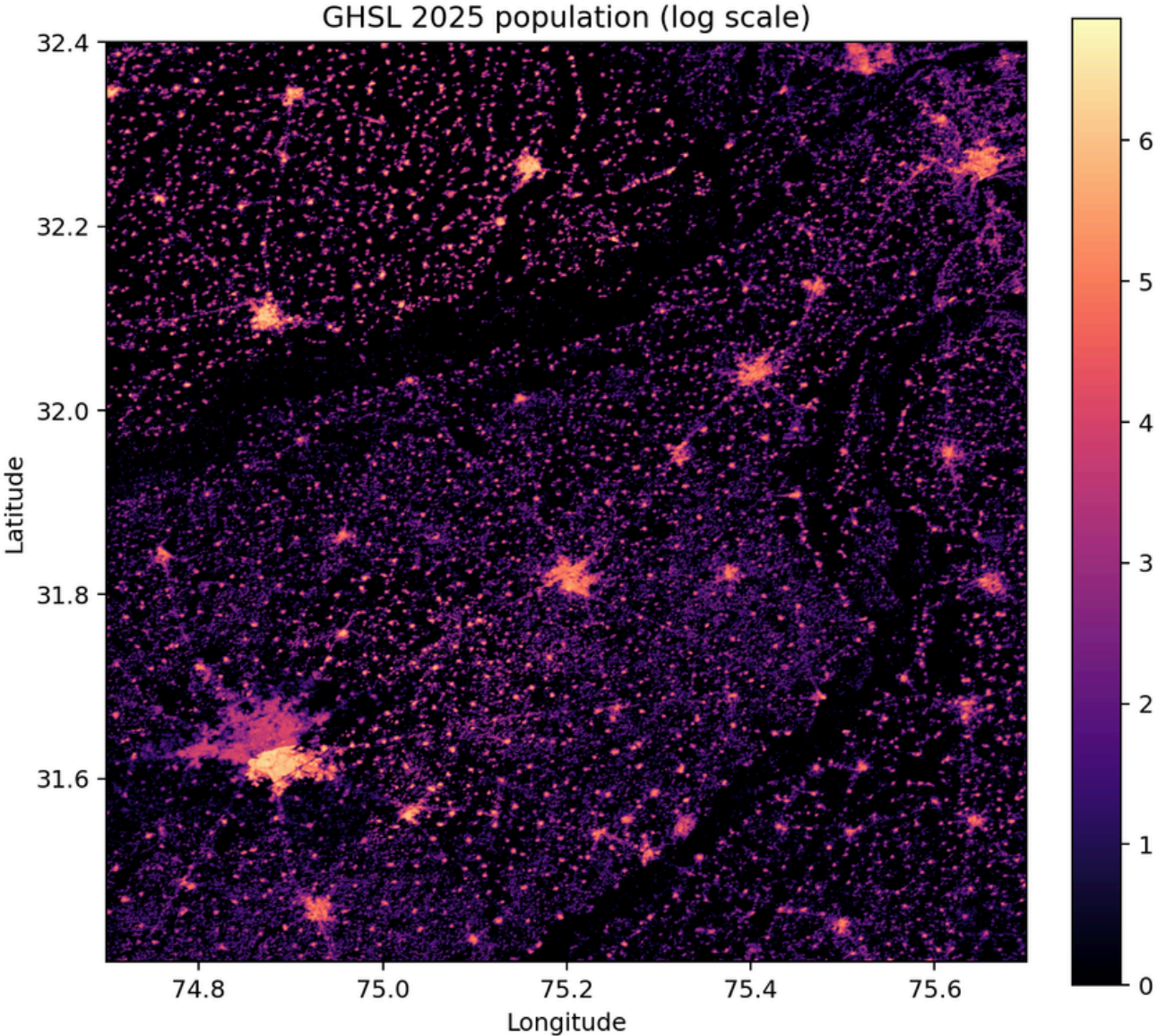


The U-Net model uses Sentinel-1 SAR imagery to detect floodwater even during cloudy weather. It achieved 90% Precision and 80.9% Recall, showing strong flood-segmentation performance.

Objective 2 Working

Input: Sentinel satellite imagery and Copernicus DEM elevation data for Amritsar.
Output: Flood mask, detected flood clusters, depth estimation, and priority classification for affected areas.

The map shows the distribution of population across the study area using GHSL 2025 data on a logarithmic scale. Brighter regions represent higher population density, highlighting communities likely to face greater flood impact. We will extend the GHSL population layer in the priority-scoring process. Areas with higher affected population and population density will receive higher priority scores



Cluster	Flood pixels	Relative area	Mean depth (m)	Priority
C-1	52,374	35.52%	1.08	CRITICAL
C-2	878	0.60%	0.18	LOW
C-3	109	0.07%	0.39	LOW
C-4	107	0.07%	0.03	LOW
C-5	95	0.06%	0.76	LOW
C-6	78	0.05%	0.04	LOW
C-7	70	0.05%	1.56	HIGH
C-8	56	0.04%	0.25	LOW

CONTRIBUTIONS ACCORDING TO CURRENT IMPLEMENTATION

KANISHKA RANI



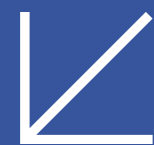
- Collected and prepared Sentinel-1/Sentinel-2 satellite data for training Models, including preprocessing and geospatial alignment.

VIVEK DUTT SHARMA



- Collected Punjab District Sentinel-1/2 images from Google EarthEngine and tested the flood-detection workflow using the U-Net/Prithvi-style model pipeline to generate flood predictions.

RITIGYA SINGH



- Converted binary flood masks and implemented connected-component analysis to identify flood clusters and assign priority to critical areas.

PIYUSH GARG



- Collected GHSL population grid of Punjab and built the dashboard interface to display input satellite model diagnostics, predicted flood masks, and flood-cluster summaries.

Current Progress

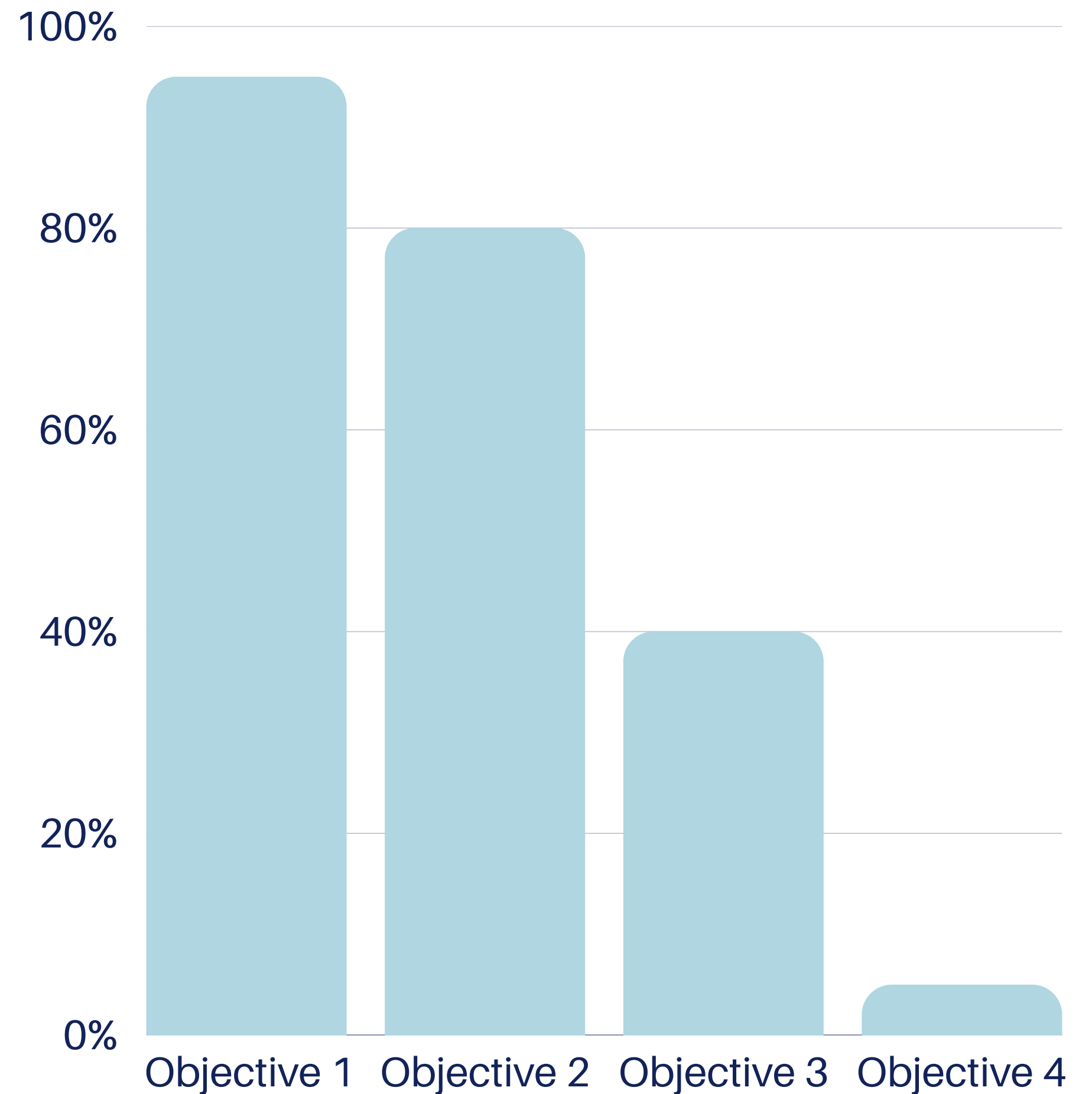
In terms of number of approved objectives

Objective 1 : To develop and train the UNet and IBM-NASA Prithvi models to accurately detect flooded areas from satellite images and generate flood masks

Objective 2 : To estimate the number of people affected by the flood by combining the generated flood masks with population and district boundary data.

Objective 3 : To calculate flood severity and assign priority scores to districts so that the most affected areas can be identified for rescue and relief operations.

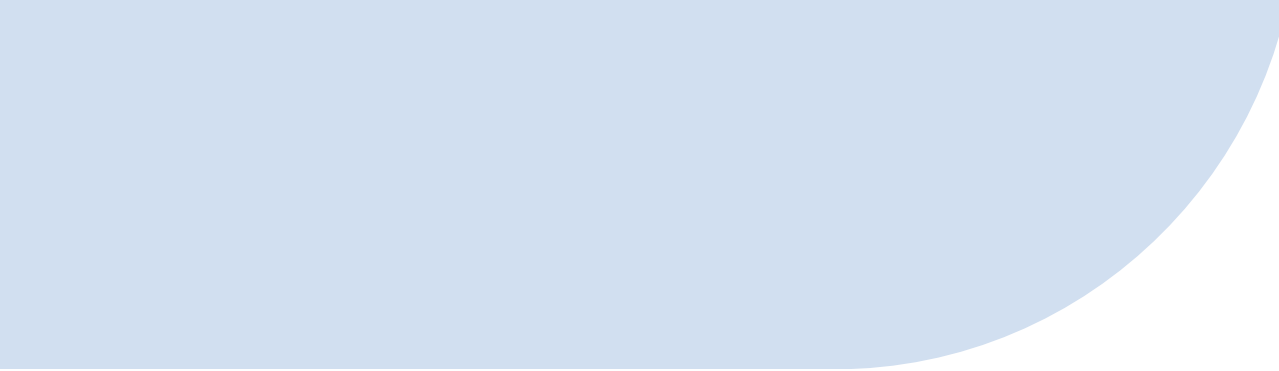
Objective 4 : To develop an interactive dashboard that displays flood maps, affected population, priority districts, and optimized rescue routes to support disaster management and decision-making.



Future Plan Of Remaining Objectives

- Complete end-to-end validation: calculate cluster-level population impact, depth/severity, road accessibility, and full six-factor priority scores.
- Run priority-aware vehicle-route optimisation on safe road networks and generate relief mission sheets for available vehicles.
- Integrate Punjab 2025 satellite information to run the implemented pipeline on Amritsar and Gurdaspur flood scenarios.
- Develop the dashboard to visualise flood clusters, critical areas, blocked roads, safe routes, and relief plans.





THANK YOU!

